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CUET UG

Biology

Biodiversity and Conservation

Official Syllabus Unit X · Class 12 · CUET UG 2027 Cycle Notes

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Biology syllabus 2026, the current 2027 baseline, Unit X)	Skills the questions test
Biodiversity — concept, patterns, importance	Biodiversity – concept, patterns and importance	Naming the three levels with everyday examples; reading the latitudinal gradient and the species–area equation.
Causes of biodiversity loss	Causes of biodiversity loss	Matching each named example (Nile perch, Steller's sea cow, Amazon clearing) to its driver.
Conservation approaches	Biodiversity conservation – approaches (narrowly utilitarian, broadly utilitarian and ethical)	Sorting a given statement into the narrowly utilitarian, broadly utilitarian or ethical view.
In situ methods	Methods of biodiversity conservation – in situ (endemism, hotspots and sacred groves etc.)	Applying the hotspot criteria (endemism + habitat loss), placing India's three hotspots, classifying sacred groves.
Ex situ methods	Methods of biodiversity conservation – ex situ (Wildlife safaris, zoological parks and cryopreservation etc.)	Mapping zoo, safari park, seed bank, tissue culture and cryopreservation to the ex situ label — the syllabus lists "Wildlife safaris" in this captive-care context.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Biology paper maps strictly to NCERT Class 12, and Unit X closes the Biology syllabus with ecology's big-picture questions: why life's variety matters and how we keep it.

What Biodiversity Really Means

Walk from a city park into a tropical forest and the world changes fast. More kinds of trees, more insects, more birds — and the variety itself has a name. The biologist Edward Wilson popularised the word **biodiversity** for the combined diversity — the total variety of life — at all levels of biological organisation, from genes up to whole ecosystems. That word packs three levels into one idea, and every exam question starts by asking which level is being discussed. So let us meet the levels one by one, with everyday examples.

- **Genetic diversity — variety inside one species:** members of the same species carry slightly different gene versions, like students of one class with different handwriting. India has more than **50,000 genetically different strains of rice** and about **1,000 varieties of mango** — all one species, many gene combinations.
- **Species diversity — variety of species in a place:** this is the level you picture first. The **Western Ghats** hold a greater amphibian species diversity (more kinds of frogs and their relatives) than the **Eastern Ghats**.
- **Ecological diversity — variety of ecosystems in a region:** an ecosystem is a community of organisms plus the physical setting they live in. India spans **deserts, rain forests, mangroves, coral reefs, wetlands, estuaries and alpine meadows** — a far greater ecosystem variety than a Scandinavian country like **Norway**.

Level	What it means, in one line	The classic example
Genetic diversity	Different gene versions **within** one species.	50,000+ rice strains and 1,000 mango varieties in India.
Species diversity	Different **species** living in one region.	Western Ghats > Eastern Ghats for amphibian species.
Ecological diversity	Different **ecosystems** across a large area.	India's deserts to coral reefs vs Norway's narrower range.

How much life have we actually counted?

- **Counted so far:** about **1.7–1.8 million species** of plants and animals have been described — formally named, checked and catalogued.
- **The real number is far higher:** the ecologist **Robert May** places global species diversity at about **7 million** — so most species on Earth are still unnamed.
- **Insects rule the count:** among animals, insects make up **more than 70 per cent** of all species — 7 out of every 10 animals on Earth is an insect.
- **A fungi surprise:** the number of fungi species in the world is **more than** the combined total of fishes, amphibians, reptiles and mammals.
- **Why counting matters:** you cannot protect what you have not named. Species lists are the base of every conservation plan.

ANALOGY — THE LIBRARY OF LIFE

Picture Earth as one giant library. Each **word** is a gene — a small unit that means little alone but combines into sense. Each finished **book** is a species — one complete working set of words. Each **shelf** is an ecosystem — books arranged, interacting, in their setting. The mapping is exact: genes = words, species = books, ecosystems = shelves. Now the hard part: we have catalogued only a fraction of the books, and we are burning shelves faster than we can read them. Most fires destroy books nobody has opened yet — species going extinct before anyone names them.

Which level is which? A 60-second drill

Questions rarely say "genetic diversity" directly. They give you an example and expect you to name the level. Practise the translation now.

WORKED EXAMPLE — NAME THE LEVEL

- **"India has more than 50,000 genetically different strains of rice."** All strains are one species, so the variety sits **within** the species → **genetic diversity**.
- **"The Western Ghats have a greater amphibian diversity than the Eastern Ghats."** Different frog species are being counted → **species diversity**.
- **"India has deserts, rain forests, mangroves, coral reefs and alpine meadows."** Different ecosystems, not different genes or species → **ecological diversity**.

The sorting rule: variety **within** one species = genetic; variety **among** species = species; variety **of habitats and ecosystems** = ecological.

- **India punches above its weight:** with only **2.4 per cent** of the world's land area, India holds about **8.1 per cent** of the world's species — a "mega-biodiversity" country in official listings.
- **Keep the three numbers together:** 1.7–1.8 million described, 7 million estimated, 70 per cent insects. They anchor many one-mark checks.

Patterns in Biodiversity

Where on Earth would you find the most species? The answer is remarkably tidy: richness climbs towards the equator and falls towards the poles, and within any region it grows with the area you explore. Two patterns — the **latitudinal gradient** and the **species–area relationship** — carry almost every question from this part of the unit.

The latitudinal gradient: the tropics win

Latitudinal gradient means the change in species richness as you move from the poles (high latitudes) to the equator (low latitudes). The tropics — the belt between **23.5° N** and **23.5° S** — harbour more species than temperate or polar zones.

- **The bird numbers tell the story: Colombia**, near the equator, has nearly **1,400 bird species**; **New York** at 41° N has **105**; **Greenland** at 71° N has only **56**.

- **India sits in the rich belt:** with much of its land in the tropics, India has **more than 1,200 bird species**.
- **Plants follow the same slope:** a tropical forest in Ecuador can hold up to **10 times** as many vascular plant species as an equal-area temperate forest in the Midwestern USA.
- **The champion region:** the **Amazonian rain forest** of South America carries the **greatest biodiversity on Earth**.

Place	Latitude	Bird species
Colombia	near 0° (equator)	nearly 1,400
India (tropical latitudes)	mostly 8–30° N	more than 1,200
New York (USA)	41° N	105
Greenland	71° N	56

Why the tropics are so rich — the three reasons

NCERT gives exactly three reasons; learn them as a fixed trio.

- **Long evolutionary time:** temperate zones were hit by repeated **glaciations** (ice ages) that reset life; the tropics stayed **relatively undisturbed for millions of years**, so species had a long run to diversify.
- **Constant climate:** tropical environments are **less seasonal and more predictable**, so species can specialise tightly into narrow niches — a **niche** is a species' job and address in its ecosystem.
- **More solar energy:** the tropics receive **more sunlight**, giving higher productivity (more plant growth), which supports more species at every level.

The species–area relationship

Count a small patch of forest, then a bigger one: the bigger patch holds more species — but the rise flattens out. The German naturalist **Alexander von Humboldt** noticed that richness grows with explored area **only up to a limit**, and the pattern fits a neat power equation.

SPECIES–AREA RELATIONSHIP

$$S = C \cdot A^{z}$$

$$\log S = \log C + z \log A$$

S = number of species

A = area explored

C = constant (intercept — species at unit area)

z = slope (how fast richness rises with area)

The second form is just the same equation with logarithms (log = the power to which 10 must be raised), which turns the curve into a straight line — that is the line whose slope is z.

- **Small and medium regions:** whether you count plants in Britain, birds in California or molluscs in New York State, the slope z falls in a narrow **0.1 to 0.2** range.
- **Very large regions:** for areas the size of **whole continents**, z steepens to **0.6 to 1.2**. For fruit-eating birds and mammals across the tropical forests of different continents, z is about **1.1**.
- **Why z matters:** a steeper slope says richness depends strongly on area — so destroying habitat there wipes out species much faster.

WORKED EXAMPLE — THE EQUATION AT WORK

A forest study fits the equation with $\mathbf{C = 10}$ and $\mathbf{z = 0.2}$. Note that $0.2 = \frac{1}{5}$, so $A^{0.2}$ is simply the **fifth root** of the area.

- **Patch A — 100 km²:** $S = 10 \times 100^{0.2}$

  $100^{0.2} \approx 2.512$

  $S \approx 10 \times 2.512 \approx 25$ species
- **Patch B — 400 km² (four times the area):** $S = 10 \times 400^{0.2}$

  $400^{0.2} \approx 3.314$

  $S \approx 10 \times 3.314 \approx 33$ species
- **Read the lesson:** four times the area lifted richness only from 25 to 33. Richness grows gently with area — and the reverse also holds gently at first, which is why the **last** fragments of a destroyed forest matter so much: the curve steepens as regions grow large (z up to 1.2).

- **Bridge to conservation:** the same equation lets biologists **predict** how many species a shrinking habitat will lose — the mathematical heart of the habitat-loss story you will meet in a later section.

Why Biodiversity Matters — Three Views

Why should a farmer, a doctor and a monk all care about a beetle? Because each looks at nature through a different window, and every window is a valid answer to "why conserve?". The syllabus groups the arguments into three views — **narrowly utilitarian**, **broadly utilitarian** and **ethical**. Learn the labels, because matching a statement to its view is a classic question type.

Narrowly utilitarian — biodiversity pays the bills

Narrowly utilitarian means valuing nature for the direct products humans extract and sell or use.

- **Food, fibre and fuel:** crops, fruits, meat, wool, cotton, firewood — the oldest "services" biodiversity offers.
- **Medicines:** about **25 per cent** of the drugs sold worldwide contain plant extracts or active ingredients derived from plants.
- **Bioprospecting:** exploring molecular, genetic and species-level diversity to find products of economic importance — nature treated as a research mine.

Broadly utilitarian — nature works for free

Broadly utilitarian means valuing the **ecosystem services** — the unpaid work nature does for everyone, whether or not anyone pays for it.

- **Oxygen:** the Amazonian rain forest is called the **"lungs of the planet"** — it contributes about **20 per cent** of the oxygen in Earth's atmosphere through photosynthesis.
- **Pollination:** bees, bumblebees, birds and bats pollinate crops; without them there would be few fruits and seeds.
- **Climate and flood control:** forests moderate rainfall, hold soil and soften floods.
- **The price tag:** in 1997, **Robert Constanza** and colleagues valued nature's life-support services at about **US $33 trillion a year** — nearly twice the global GNP (US $18 trillion).
- **NCERT's key point:** a single species' product value can be tiny compared with its **service** value — a bee costs almost nothing in the market, yet its pollination work keeps farms alive.

Ethical — every species has intrinsic value

Ethical means valuing species for their own sake — the philosophical window.

- **Intrinsic value:** every species has worth **in itself**, even if it earns humans nothing today.
- **A duty to the future:** we should pass biological wealth on to coming generations in as good a shape as we received it.
- **People already act on it:** the **Bishnoi** community of Rajasthan protects trees and wild animals as a religious duty.

View	Core idea, in one line	One example each
Narrowly utilitarian	Nature supplies direct products we use or sell.	About 25% of world drugs carry plant-derived ingredients.
Broadly utilitarian	Nature runs free ecosystem services for all.	Amazon ≈ 20% of atmospheric oxygen; services ≈ US $33 trillion a year.
Ethical	Every species has intrinsic value and moral worth.	Bishnoi villagers shield khejri trees and blackbuck.

ANALOGY — THE UTILITY COMPANY WITH NO INVOICE

Imagine a company that fills your oxygen cylinder, pollinates your food crops, recycles your water and cools your city — and never sends a bill. That company is biodiversity. The **narrowly utilitarian** customer buys its products at the counter; the **broadly utilitarian** customer values the unpaid services and refuses to let the company shut down; the **ethical** citizen insists every employee — every species, however humble — deserves to keep its job. A conservation question is simply asking you: which customer is speaking?

Causes of Biodiversity Loss — The Four Big Drivers

Species rarely vanish for a mystery reason. The ecologist Jared Diamond grouped the four biggest human-made causes as the **"Evil Quartet"**, and NCERT keeps the same four. For each driver below, fix three things in memory: what it is, one named example, and why it hurts. Most questions hand you the example and expect the driver.

1. Habitat loss and fragmentation — the biggest cause

- **What it is:** natural habitats are destroyed outright, or chopped into small, disconnected pieces (**fragmentation**).
- **Named example:** the **Amazon rain forest** is being cut and cleared for **soybean cultivation** and for **grasslands to raise beef cattle**.
- **The shrinking numbers:** tropical rain forests once covered more than **14 per cent** of Earth's land surface; barely **6 per cent** remains.
- **Why fragmentation hurts:** large mammals need big territories and migratory birds need connected routes; small, broken patches can feed neither, so their populations decline.

2. Over-exploitation — harvesting faster than nature refills

- **What it is:** humans take a species — hunted, fished, collected — faster than it can breed and replace itself.
- **Named examples: Steller's sea cow** (a large sea mammal) and the **passenger pigeon** were once abundant; both were driven **extinct** by overhunting. Marine fish stocks crash under the same pressure.
- **Why it hurts:** slow-breeding species cannot recover between harvests, so each season's catch eats into the breeding stock itself.

3. Alien species invasions — new arrivals, old losses

- **What it is:** species moved by humans outside their natural range (**alien** species) sometimes turn **invasive** — spreading fast and crowding out native life.
- **Named example:** the **Nile perch**, introduced into **Lake Victoria** in East Africa, wiped out **more than 200 species of cichlid fish** — the lake's native fish family.
- **India's invaders:** the weeds **Parthenium** (carrot grass), **Lantana** and **water hyacinth** (**Eichhornia**) damage land and freshwater habitats alike.
- **One more for the list:** the **African catfish** (**Clarias gariepinus**), brought in for aquaculture, now threatens India's native catfishes; in Australia, introduced predators such as feral cats have helped push many small native mammals extinct.
- **Why it hurts:** natives evolved without the new arrival's weapons, so they are outcompeted or eaten — no escape route exists.

4. Co-extinctions — when partners fall together

- **What it is:** when one species goes extinct, species **obligately tied** to it — partners that cannot live without it — go extinct too.
- **Named example:** when a **host fish** species dies out, its unique assemblage (set) of **parasites** dies with it; a coevolved plant and its **pollinator** face the same chain reaction.

- **Why it hurts:** extinction never travels alone — it always collects dependents on the way down.

Driver	Flagship example	One-line effect
Habitat loss & fragmentation	Amazon cleared for soybeans and cattle grasslands	Rain forest down from 14% to barely 6% of land area.
Over-exploitation	Steller's sea cow; passenger pigeon	Harvested faster than they could breed — extinct.
Alien species invasion	Nile perch in Lake Victoria	More than 200 cichlid fish species wiped out.
Co-extinction	Host fish and its unique parasites	Obligate partners die along with the host.

QUICK DRILL — MATCH THE EXAMPLE TO ITS DRIVER

- **"Tropical rain forests fell from 14% to 6% of the land."** → **habitat loss** — the habitat itself was removed.
- **"Steller's sea cow disappeared soon after hunters found it."** → **over-exploitation** — taken faster than it bred.
- **"Water hyacinth chokes Indian ponds."** → **alien species invasion** — an introduced plant turning invasive.
- **"A fish host dies and its parasites die too."** → **co-extinction** — an obligate partner falls with its host.

The sorting question to ask: **did humans remove the home, take the species, bring a newcomer, or did a partner drag it down?**

Conserving Biodiversity — In Situ

Diagnosis done; now the treatment. **In situ conservation** (Latin for "in the original place") means protecting a species **inside its natural home** — the habitat is fenced, guarded and left working. It is the first choice of conservation because saving a habitat saves **every species in it at once**, not one species at a time.

The in situ toolbox

Tool	What it is
National park	A strictly protected natural area — habitat and wildlife are fully preserved.
Wildlife sanctuary	A protected area reserved for animals, often centred on particular threatened species.

Tool	What it is
Biosphere reserve	A large area conserving whole ecosystems, while allowing research and limited human activity.
Sacred grove	A forest patch protected by a local community for religious or cultural reasons.

Sacred groves — conservation by faith

- **What they are:** tracts of forest set aside and **worshipped** by villages; every tree and animal inside gets total protection.
- **Khasi and Jaintia Hills (Meghalaya):** famous sacred groves, kept untouched across generations.
- **Bishnoi villages (Rajasthan):** the community protects **khejri trees** and **blackbuck** as a religious duty; **Amrita Devi Bishnoi** and 363 others gave their lives defending khejri trees.
- **The exam-relevant label:** sacred groves are **in situ** — the forest is protected exactly where it stands, never moved.

Hotspots — where scarce money should go first

- **Endemism first: endemic** species are those found **nowhere else on Earth** — lose them here and they are lost from the planet.
- **Hotspot definition:** a **biodiversity hotspot** is a region with **very high species richness**, **very high endemism and** very high habitat loss. Both halves matter; a rich but safe region is not a hotspot.
- **India's three hotspot regions:** the **Western Ghats–Sri Lanka** region, the **Himalaya**, and **Indo-Burma**.
- **Why hotspots guide policy:** money and protection laws cannot cover every hectare, so the places with the most unique, most threatened life come first.

Conserving Biodiversity — Ex Situ

Sometimes the home is already gone, or so few animals remain that the wild cannot carry them. Then conservation moves **ex situ** (Latin for "away from the original place"): the threatened species — or its seeds, gametes and tissues — is protected **outside its natural habitat, under human care**. Figure 1 lines the two toolkits up side by side — read it once here, and once at revision time.

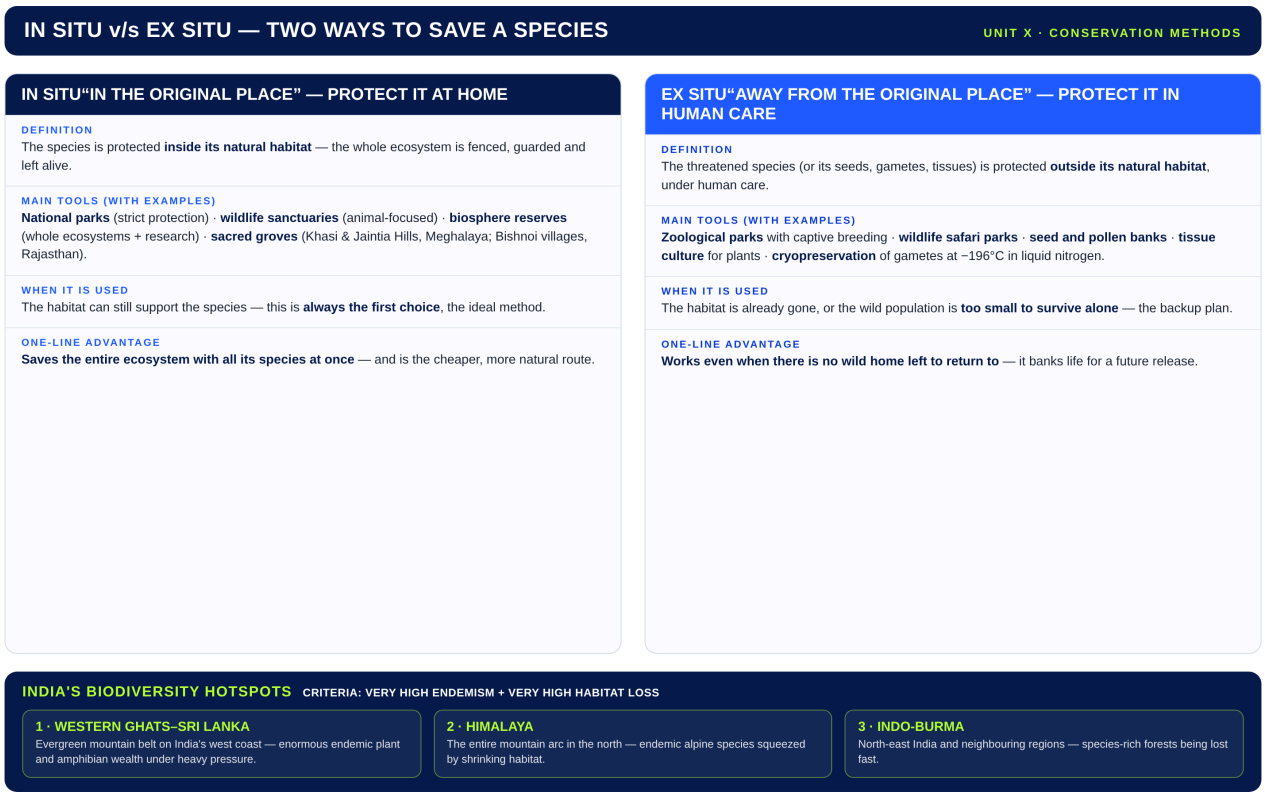


Figure 1: In situ vs ex situ conservation — the two methods side by side, with India's three hotspot regions in the closing strip.

The ex situ toolbox

Tool	What it keeps, and how
Zoological park (zoo)	Live animals kept and bred in captivity — **captive breeding** raises numbers for a guarded future.
Wildlife safari park	Animals roam large open enclosures and are viewed from vehicles — the syllabus lists “Wildlife safaris” among ex situ methods.
Seed / pollen bank	Seeds and pollen of threatened plants stored dry and cold for long periods.
Tissue culture	Whole plants regrown from small tissue pieces on nutrient jelly in the laboratory.
Cryopreservation	Gametes (sperm and egg cells) of threatened animals frozen in **liquid nitrogen at −196 °C**, staying viable and fertile for years.

- **In vitro fertilisation:** eggs can also be fertilised in the laboratory, adding another ex situ route for threatened animals.
- **When ex situ is the only option:** when the habitat has been destroyed, when the wild population has crashed below a self-sustaining level, or when the species must be banked until a safe home can be restored.

- **The honest limit:** ex situ saves the species' genes and bodies, but not the ecosystem it belonged to — that is why in situ remains the first choice and ex situ the safety net.

ANALOGY — THE BACKUP DRIVE

A seed bank, a zoo breeding programme and a frozen gamete store are life's **backup drive**. You copy your important files somewhere safe and hope never to need the copies; if the original computer (the wild habitat and its population) fails, the backup can restore the work. The mapping is exact: seeds and gametes = files, banks and zoos = drives, and a future reintroduction = restoring the backup onto a repaired machine. No backup, no second chance.

The Whole Picture — Revision Flow

Read the chapter as one connected story and it stops being ten lists. You define the variety, measure it, price it, diagnose what is destroying it, and then choose the protection tool that fits. That is also the order questions travel in.

- **Step 1 — Define it:** biodiversity is the combined variety of life at three levels — **genetic, species, ecological** — a word popularised by Edward Wilson.
- **Step 2 — Measure it:** about **1.7–1.8 million** species described, maybe **7 million** exist; richness rises towards the tropics (**latitudinal gradient**) and with area (**S = C dot A^z**).
- **Step 3 — Value it: narrowly** utilitarian (products), **broadly** utilitarian (ecosystem services), **ethical** (intrinsic value).
- **Step 4 — Diagnose it:** the four drivers — **habitat loss and fragmentation, over-exploitation, alien invasions, co-extinctions**.
- **Step 5 — Protect it: in situ** first (parks, sanctuaries, reserves, sacred groves; hotspots guide priority), **ex situ** as the backup (zoos, safari parks, seed banks, tissue culture, cryopreservation).

How to think about a conservation MCQ

- **Name the level first:** one species, many varieties → genetic; many species → species; many ecosystems → ecological.
- **Pin every named example to its driver:** Nile perch → alien invasion; Steller's sea cow → over-exploitation; Amazon clearing → habitat loss.
- **Test both hotspot criteria:** very high endemism **and** very high habitat loss — India's three are Western Ghats–Sri Lanka, Himalaya, Indo-Burma.
- **Sort the method before the option list:** sacred grove → in situ; zoo, seed bank, −196 °C store → ex situ.
- **Handle the species–area numeric:** use the log form, know that z sits at 0.1–0.2 for small regions and 0.6–1.2 for continents, and remember $A^{0.2}$ is the fifth root of A.
- **Attach the fixed numbers:** 25% of drugs from plants; Amazon ≈ 20% of atmospheric oxygen; rain forest 14% → 6%; services ≈ US $33 trillion a year.

Clue in the question	Concept to recall
"Found nowhere else on Earth"	Endemism — the first half of the hotspot criteria
"More than 200 cichlids disappeared"	Alien species invasion — the Nile perch
"A price tag on nature's services"	Broadly utilitarian view — ecosystem services, US $33 trillion
"z between 0.1 and 0.2"	Species–area slope for small and medium regions
"A forest patch worshipped by a village"	Sacred grove — in situ, community-protected
"Gametes stored at −196 °C"	Cryopreservation — an ex situ method

Key Facts & Terms Cheatsheet

KEY FACTS & TERMS CHEATSHEET — UNIT X

Term / fact	Meaning	Example / number
Genetic diversity	Variety of genes **within** one species	50,000+ rice strains; 1,000 mango varieties (India)
Species diversity	Variety of species in a region	Western Ghats > Eastern Ghats for amphibians
Ecological diversity	Variety of ecosystems across an area	India's deserts to coral reefs vs Norway's narrower set
Species–area equation	$S = C \cdot A^z$ log form: $\log S = \log C + z \log A$	Bigger area → more species, up to a limit
Letters of the equation	S = species count, A = area C = intercept constant, z = slope of the line	z is the slope of the log-form straight line
z, small and medium regions	Slope for plants in Britain, birds in California, molluscs in New York	0.1 to 0.2
z, very large regions	Slope for whole continents; fruit-eating birds and mammals	0.6 to 1.2 (frugivores ≈ 1.1)
Latitudinal gradient	Richness climbs towards the tropics (23.5° N to 23.5° S)	Colombia 1,400 birds vs Greenland 56

Term / fact	Meaning	Example / number
Hotspot criteria	Very high endemism + very high habitat loss — both needed	India's 3: Western Ghats–Sri Lanka, Himalaya, Indo-Burma
Endemism	Species found nowhere else on Earth	Western Ghats' endemic amphibians
Habitat loss & fragmentation	Homes destroyed or split into small patches — the biggest driver	Amazon for soy and cattle; rain forest 14% → 6%
Over-exploitation	Harvested faster than the species can breed	Steller's sea cow; passenger pigeon — extinct
Alien species invasion	Introduced species spreads and outcompetes natives	Nile perch → 200+ cichlids gone; Lantana; water hyacinth; Parthenium
Co-extinction	Obligate partners die when their host dies	Host fish and its unique parasite set
In situ, in one line	Protect the species inside its natural home	National parks, sanctuaries, biosphere reserves, sacred groves
Ex situ, in one line	Protect the species away from its natural home	Zoos (captive breeding), safari parks, seed banks, tissue culture, cryopreservation
Sacred groves	Community-protected forest patches — in situ by faith	Khasi & Jaintia Hills (Meghalaya); Bishnoi villages (Rajasthan)
Cryopreservation	Gametes of threatened species kept viable and fertile in liquid nitrogen	−196 °C

Numbers to carry everywhere: 1.7–1.8 million species described; 7 million estimated (Robert May); insects > 70% of animal species; India 2.4% of land, 8.1% of species; 25% of drugs from plants; Amazon ≈ 20% of atmospheric oxygen; rain forest 14% → 6%; services ≈ US $33 trillion/yr.

Common Errors and Special Cases

COMMON ERRORS & SPECIAL CASES

The mistake students make	What is right	Why the trap works
Calling an endemic species "already extinct"	**Endemic** = found nowhere else; **extinct** = gone forever. A species can be both only after it disappears.	Both words describe range and survival, so they sound similar until you pin each definition.
Treating a hotspot as simply "a place with many species"	A hotspot needs very high endemism **and** very high habitat loss — both criteria together.	Richness alone misses the official definition; the threat half is what makes protection urgent.
Listing the Nile perch under over-exploitation	The Nile perch episode is **alien species invasion** — the fish was introduced into Lake Victoria.	A fish dying looks like fishing; the species was never hunted, it **arrived** and outcompeted natives.
Treating sacred groves as ex situ, or as tourist sites	Sacred groves are **in situ** — the forest is protected exactly where it stands.	"Religious" makes students think of people, not habitat; nothing is ever moved.
Writing co-extinction for any species that dies out	Co-extinction needs an **obligate partner** — a host–parasite or plant–pollinator pair.	The word only applies when a second species is dragged down by the first.
Applying $S = C \cdot A^z$ without the log form	The slope z comes from the straight-line log form: $\log S = \log C + z \log A$.	The power equation is a curve; only its logarithm draws the line whose slope is z.
Assuming ex situ means releasing animals into the wild	Ex situ keeps the species **away** from its natural home, under human care; release is a later step, not the method.	Students picture freedom; the defining feature of ex situ is exactly the opposite.

- **Three checking habits:** attach every named example to its driver before reading the options; tick **both** hotspot criteria, never one; and match each number (−196 °C, 25%, 0.1–0.2, 0.6–1.2) to its term the night before the exam.

The Whole Chapter on One Page



Figure 2: Whole chapter on one page — the three levels, the two patterns, the three views, the four drivers, the two conservation toolkits, and the traps that catch most students.

Practice Questions

Q1. The word **biodiversity** — the combined diversity at all levels of biological organisation — was popularised by:

(A) Robert May
(B) Robert Constanza
(C) Alexander von Humboldt
(D) Edward Wilson

Q2. India has more than 50,000 genetically different strains of rice and about 1,000 varieties of mango. This variety **within** a single species is an example of:

(A) genetic diversity
(B) species diversity
(C) ecological diversity
(D) habitat fragmentation

Q3. In the species–area relationship $S = C \cdot A^z$, a region has $C = 10$ and $z = 0.2$. For an area of 10,000 km², the predicted number of species is closest to:

(A) 20
(B) 63
(C) 100
(D) 200

Q4. Colombia near the equator holds nearly 1,400 bird species, while Greenland at 71° N holds only 56. The steady fall in richness from the tropics towards the poles is called the:

(A) species–area relationship
(B) niche specialisation
(C) latitudinal gradient
(D) habitat fragmentation

Q5. Exploring molecular, genetic and species-level diversity to find products of economic importance — for example, new medicines from plants — is called:

(A) biopiracy
(B) bioremediation
(C) biomass estimation
(D) bioprospecting

Q6. For very large areas such as entire continents, the slope z of the species–area relationship is found to lie close to:

(A) 0.1 to 0.2
(B) 0.2 to 0.4
(C) 0.6 to 1.2
(D) 1.5 to 2.0

Q7. The Nile perch, introduced into Lake Victoria, wiped out more than 200 species of cichlid fish. This loss is best classified as caused by:

(A) over-exploitation
(B) alien species invasion
(C) habitat fragmentation
(D) co-extinction

Q8. Steller's sea cow and the passenger pigeon disappeared soon after humans began hunting them heavily. The driver of their extinction was:

(A) alien species invasion
(B) co-extinction
(C) over-exploitation
(D) habitat fragmentation

Q9. A region qualifies as a biodiversity hotspot only when it has:

(A) very high endemism together with very high habitat loss
(B) the largest number of national parks in the country
(C) the world's largest area of tropical forest
(D) a government ban on all hunting and fishing

Q10. Gametes of a threatened animal are kept viable and fertile for long periods in liquid nitrogen at −196 °C. This ex situ technique is called:

(A) tissue culture
(B) cryopreservation
(C) in situ conservation
(D) deforestation control

Solutions

Sol. Q1. **Biodiversity** as the combined diversity at all levels of biological organisation was popularised by the biologist **Edward Wilson**. Robert May estimated 7 million species; Robert Constanza priced ecosystem services; Alexander von Humboldt gave the species–area idea.

Answer: (D)

Sol. Q2. All rice strains and mango varieties belong to **one species**; the differences lie in gene versions **within** that species → **genetic diversity**. Species diversity would need many different species, ecological diversity many different ecosystems.

Answer: (A)

Sol. Q3. Since $0.2 = \frac{1}{5}$, the factor $10000^{0.2}$ is the fifth root of 10,000, which is **6.31**. Then $S = 10 \times 6.31 \approx 63$. (Check: $6.31^5 \approx 10,000$.)

Answer: (B)

Sol. Q4. Richness falling steadily from equator to pole — Colombia 1,400, New York 105, Greenland 56 — is the **latitudinal gradient**: the tropics (23.5° N to 23.5° S) hold the most species. The species–area relationship instead links richness to **area**, not latitude.

Answer: (C)

Sol. Q5. Systematically searching biodiversity at the molecular, genetic and species level for economically useful products is **bioprospecting**. Bioremediation is cleanup with organisms; biopiracy is unauthorised commercial use of traditional knowledge.

Answer: (D)

Sol. Q6. For small and medium regions z sits at 0.1–0.2, but for **very large areas** — whole continents — the slope steepens to **0.6 to 1.2**; for fruit-eating birds and mammals across tropical forests of different continents it is about 1.1.

Answer: (C)

Sol. Q7. The Nile perch was **introduced** into Lake Victoria and then outcompeted the native cichlids — no hunting of cichlids, no removal of their habitat. An introduced species turning invasive = **alien species invasion**.

Answer: (B)

Sol. Q8. Both species were abundant until humans harvested them faster than they could breed — the definition of **over-exploitation**. No new species arrived and no obligate partner was involved.

Answer: (C)

Sol. Q9. The hotspot label carries **both** official criteria: very high species richness with very high **endemism**, **plus** very high habitat loss. Park counts, forest size or legal bans are not part of the definition.

Answer: (A)

Sol. Q10. Freezing gametes in liquid nitrogen at **−196 °C** so they stay viable and fertile for long periods is **cryopreservation** — an ex situ method, because the material is preserved away from the natural setting. Tissue culture regrows plants; in situ protects species in the wild.

Answer: (B)

FINAL DRILL — SAY EACH ANSWER BACK IN ONE LINE

Cover the right half, say the recall aloud, then check yourself.

Prompt	One-line recall
Three levels of biodiversity?	Genetic (within species), species, ecological (ecosystems).
Species–area equation and its slope?	$S = C \cdot A^z$; log form is a straight line with slope z (0.1–0.2 small, 0.6–1.2 continental).
Three views on why biodiversity matters?	Narrowly utilitarian products, broadly utilitarian services, ethical intrinsic value.
Four drivers of loss, with one flagship each?	Habitat loss (Amazon), over-exploitation (Steller's sea cow), alien invasion (Nile perch), co-extinction (host's parasites).
Hotspot formula?	Very high endemism + very high habitat loss → Western Ghats–Sri Lanka, Himalaya, Indo-Burma.
Two conservation toolkits?	In situ at home (parks, sanctuaries, reserves, sacred groves); ex situ away (zoos, safaris, seed banks, tissue culture, −196 °C).

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